Contents

I The Rising Sea **2**

1 2.2.6. Definition: Sheaf. **2**
1.1 Example 2

2 2.2.B. Presheaves that are not SHEAVES. **2**
2.1 (a) 2
2.2 (b) 2

3 2.2.7. **2**

4 2.2.C. **3**

5 2.2.D. **3**

6 2.2.E. **3**

7 2.2.F. **3**

8 2.2.G. **3**
8.1 (a) 3

9 2.2.9. **3**

10 2.2.10. **3**
10.1 References 3
10.2 3

11 2.2.E. **4**

12 2.2.F. **5**

13 2.2.G. **5**
13.1 (a) 5
13.2 (b) 5

14 2.2.11. Espace Étalé **5**

15 2.2.H. **5**

16 2.3.A. **6**

17 2.3.B. **6**

18 2.3.C. **6**
18.1 Verification 8
18.2 Warning 8
18.3 Warning: References 8
18.4 Counterexample 8
18.5 Abelian group structure 9

19 2.3.D. **10**
19.1 (a) 10
19.2 References 10

20 2.3.E. **10**

21 2.3.G. 2.3.H. **10**

22 2.3.J. **11**
22.1 References 11

23 2.4.A. **11**

24 2.4.2. Compatible Germs **11**
24.1 Two Equivalent Definitions 11
24.2 Section induces a choice of compatible germs 12

25 2.4.B. **12**

26 2.4.C. **14**

27 2.4.D. **14**
27.1 Warning 14

28 2.4.E. **14**
28.1 (a) 14
28.2 (b) 14
28.3 (c) 15

29 2.4.O. **15**
29.1 Complex analysis background 15

30 2.5.1. **16**
30.1 Remarks 16

31 2.5.B. **16**
31.1 Another Proof 16
31.2 Remarks 16

32 2.5.D. **16**

33 2.6.A. **16**
33.1 Solution 17

34 2.7.A. **17**

35 2.7.D. **17**

36 2.7.G. **17**

37 7.2.A. **17**
37.1 Comments 19

38 7.2.D. **19**
38.1 (a) 19

39 7.2.E. **20**

40 7.3.A. **20**

41 7.3.B. **20**

42 7.3.D. **20**

43 7.3.F. **20**

44 7.3.G. **20**

45 7.3.6. **21**

46 7.3.O. **21**

47 7.3.L. **21**

48 7.4. **21**

49 7.5.B. **21**

50 7.5.4. **21**

51 14.1.C. **21**

52 14.1.D. **21**

53 14.2.A. **22**

54 14.4.D. **22**
54.1 (a) 23

55 15.2.J. **24**

56 18.5.A. **24**
56.1 (a) 24
56.2 (b) 24

57 23.1.C. **24**
57.1 References 24

58 23.4.F. **24**

Part I
The Rising Sea

1 2.2.6. Definition: Sheaf.

Comments on $\mathscr{F}(\emptyset)$. In category **Set**, the empty set is initial object and one element set is terminal. See Wiki's examples HERE.

1.1 Example

2 2.2.B. Presheaves that are not SHEAVES.

2.1 (a)

For (a): see Wiki's counterexample HERE, which gave an explanation for presheaves on $\mathbb{R}$ instead of $\mathbb{C}$. See a post HERE.

2.2 (b)

See a post HERE, HERE, and HERE.

Proof.
This problem is based on some knowledge from complex analysis. Here are some facts:

- When does a complex function have a square root?
- Theorem 6.2 on Page 100 of [**?**]. This means $f(x) = x$ as a function on $\mathbb{C}$ does not admit a square root for it will vanish.

However, we can cover $\mathbb{C}$ by two slit regions $U_1 = \mathbb{C} - (-\infty, 0], U_2 = \mathbb{C} - [0, \infty)$. And on each U_i, $f\mid_{U_i}$ admits a square root and satisfy gluability axiom.

The solution given in HERE is saying on annulus, $f(x) = x$ cannot have a square root. And then we can follow argument of HERE.

□

3 2.2.7.

See Daping's Notes Definition 2.5 on Page 4 HERE; also a post HERE; also a post HERE.

See a post HERE and HERE regarding gluing sheaves.

4 2.2.C.

Since $\cup U_i$ is colimit for $\{U_i\}$, then $\mathscr{F}(\cup U_i)$ will be limit because of the contravariance. ?

See an expository post HERE.

See another post, with more detailed explanations HERE. "In fancy language, it's stack"...

5 2.2.D.

(b) Motivating example for definition of sheaf.

6 2.2.E.

See comments for Exercise 1.1. of Chapter 2 [**?**].

7 2.2.F.

Almost by definition.

8 2.2.G.

8.1 (a)

9 2.2.9.

See a post HERE.

10 2.2.10.

10.1 References

See this regarding constant presheaf. It's different from a post HERE, and Wiki's page on Constant pre-Sheaf. Why???

10.2

Clearly it's a contravariant functor

$$\mathscr{F} := \underline{S}_{\text{pre}} : \mathbf{Top}(X) \to \mathbf{Set}$$

Let $X = \{a, b\}$ with discrete topology. Pick two sections

$$s_1 \in \mathscr{F}(\{a\}) = S,\ s_2 \in \mathscr{F}(\{b\}) = S$$

such that $s_1 \neq s_2$ given S has at least two distinct elements. Clearly we have

$$s_1 \mid_{\{a\}\cap\{b\}} = s_1 \mid_{\emptyset} = e = \cdots = s_2 \mid_{\{a\}\cap\{b\}} .$$

If it's a sheaf then there exists a global section $s \in \mathscr{F}(\{a, b\}) = S$ such that

$$s_1 = s \mid_{\{a\}} = s \mid_{\{b\}} = s_2,$$

contradiction. It follows that constant presheaf defined this way is not necessarily a sheaf.

11 2.2.E.

We have to deceptively identical pre-sheaves $\mathscr{F}_1$ defined as locally closed, and $\mathscr{F}_2$ defined by giving S discrete topology...

We wish to prove they, as pre-sheaves, are isomorphic. Equivalent, we need to exhibit a natural transformation that admits an inverse. And it suffices to prove by element inclusions:

- Let $f : U \to S$ be a map that's locally constant. Now we take $g(u) = f(u)$ as a map $g : U \to S$ with S endowed with a discrete topology. We claim that g is continuous. It suffices to check for each $s \in S$, the fiber $g^{-1}(s)$ is open. For any point $a \in g^{-1}(s) = f^{-1}(s)$, there exists an open neighborhood $V_a \subset f^{-1}(s)$ such that

  $$f(V_a) = \{s\}$$

  given f is locally constant. While $V_a \subset g^{-1}(s)$, therefore we know $g^{-1}(s)$ is open and g is continuous.

- Conversely, we assume $g : U \to S$ with S given a discrete topology is continuous. We claim $f = g$ is locally constant. For any point $p \in U$, there is an open neighborhood

  $$g^{-1}(f(p)) \ni p$$

  such that f is constant because $f(g^{-1}(f(p))) = \{p\}$.

Now we try to check constant sheaf $\mathscr{F}$ is indeed a sheaf. We're going to prove identity axiom and gluability axiom using the "better description", which is much easier to check:

- If we have two functions, which equal whenever we restric to any open subset from an open cover, then they must be equal. For functions are precisely defined this way.

- Define the global section for any choice, and it's going to be well-defined for they're compatible.

Therefore $\mathscr{F} = \underline{S}$ is indeed a sheaf.

12 2.2.F.

Same argument as 5.

13 2.2.G.

13.1 (a)

It's clearly a pre-sheaf.

Fix an open subset $U \subset X$ with an open cover $\{U_i\}_{i \in I}$ for some index set I. Denote the presheaf as $\mathscr{F}$.

Pick two continuous maps $s_1, s_2 : Y \to X$ that satisfying the requirements, i.e. $s_1, s_2 \in \mathscr{F}(U)$.

Both functions will agree on U since

$$\mathrm{Res}_{U,U_i} \ \ s_1 = \mathrm{Res}_{U,U_i} \ \ s_1$$

for arbitrary U_i, whose union is U. So we must have $s_1 = s_2$.

Again with this open cover $\{U_i\}_{i \in I}$ and $a_i \in \mathscr{F}(U_i)$ for $i \in I$. Equivalently, we know $a_i : U_i \to Y$ is a continuous map satisfying $\mu \circ a_i = \mathrm{Id}\,|_{U_i}$. Now let's define a map

$$\begin{aligned} f : U &\to Y \\ u &\mapsto a_i(u) \ \ \text{when} \ \ u \in U_i. \end{aligned}$$

It's well-defined by our assumption. Also it's continuous since preimage of an open set in $V \subset Y$ is a union of open subsets given by continuity of each a_i. Similarly we can check $\mu \circ f = \mathrm{Id}\,|_U$ as expected. Unverified ?

13.2 (b)

14 2.2.11. Espace Étalé

See a post discussion accent letter in LaTeX HERE.

See an exercise in [**?**] Chapter 2, Exercise 1.13.

See the discussion after Lemma 7. on Page 229 of [**?**].

For *section*, see Wiki's explanation for *section* in context of fiber bundle; and *section* in terms of category theory.

See a detailed post HERE.

15 2.2.H.

Clearly it's again a contravariant functor, therefore $\pi_* \mathscr{F}$ must be a pre-sheaf. When $\mathscr{F}$ is a sheaf, I checked identity axiom (lots of things to write down).

16 2.3.A.

I'm planning to use universal property to define the induced map ϕ_P.

One crutial step is to verify the diagram below is commutative

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r, "\rho_{UV}"] \arrow[d, "\phi(U)"'] & \mathscr{F}(V) \arrow[d, "\phi(V)"] \\
\mathscr{G}(U) \arrow[r, "\tau_{UV}"'] \arrow[dr] & \mathscr{G}(V) \arrow[d] \\
& \mathscr{G}_P
\end{tikzcd}$$

And this is because the square diagram in the upper half commute given ϕ is a natural transformation; the lower half is by definition of $\mathscr{G}_P$. Then by universal property of colimit induces a map

$$\phi_P : \mathscr{F}_P \to \mathscr{G}_P$$

which makes the diagram commute.

See a post defined the map HERE.

17 2.3.B.

To define a functor $\pi_* : \mathbf{Set}_X \to \mathbf{Set}_Y$. Firstly, we have to define for any $\mathscr{F} \in \mathbf{Set}_X$,

$$\pi_*(\mathscr{F})(U) = \mathscr{F}(\pi^{-1}(U))$$

for any $U \in \mathfrak{Top}(X)$ as in **??**.

Secondly, for any natural transformation $\phi : \mathscr{F} \to \mathscr{G}$, we define $\pi_*(\phi)$ by specifying

$$\pi_*(\phi)(U) \ \mapsto \mathscr{F}(\pi^{-1}(U)) \to \mathscr{G}(\pi^{-1}(U)).$$

? Is this correct

18 2.3.C.

This is Exercise 1.15. from Chapter II of [**?**] on Page 67.

Proof.
Clearly $\mathrm{Hom}(\mathscr{F}, \mathscr{G})(U)$ takes value in the set of all natural transformations from $\mathscr{F}\mid_U$ to $\mathscr{G}\mid_U$. Namely, we have

$$U \mapsto \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U).$$

The restriction map induced by $V \subset U$ is given by consider a natural morphism $\alpha \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ as a natural morphism from

$$\mathscr{F}\mid_V \to \mathscr{G}\mid_V .$$

So set-theoretically restriction map is identity map, with exception that it regard an element as a presheaf on a smaller open subset.

?

Hence $\mathrm{Hom}(\mathscr{F},\mathscr{G})(\cdot)$ is a presheaf.

Fix an open subset $U \subset X$, with an open covering $\{U_i\}_{i\in I}$ for some index set I. Pick two natural transformations $\alpha, \beta \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$. Assume for any $i \in I$,

$$\mathrm{Res}_{U,U_i}\,\alpha = \mathrm{Res}_{U,U_i}\,\beta.$$

More precisely, this means for any open subset $V_i \subset U_i$ where $i \in I$ is arbitrary, we have

$$\alpha\mid_{U_i}(V_i) = \beta\mid_{U_i}(V_i).$$

However, note $\{U_i\}_{i\in I}$ is an open cover for the whole space U we're considering. It follows that for any open subset $W \subset U$, we can denote $W_i = W \cap U_i$ and express W as a union of W_i where $i \in I$.

$$\begin{aligned}
\alpha(W) =& \alpha\left(\bigcup_{i\in I} W_i\right) \in \mathrm{Obj}(\mathbf{Set}) \\
=& \bigcup_{i\in I} \alpha(W_i) \\
=& \bigcup_{i\in I} \alpha\mid_{U_i}(W_i) \\
=& \bigcup_{i\in I} \beta\mid_{U_i}(W_i) \\
=& \cdots \\
=& \beta(W).
\end{aligned}$$

Here the third equality holds by the definition of restriction map. While $W \subset U$ is arbitrary, it follows that $\alpha = \beta \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ as expected.

It remains to check gluability. Again $\{U_i\}_{i\in I}$ is an open cover of U. Pick natural transformations $\alpha_i \in \mathrm{Mor}(\mathscr{F}\mid_{U_i}, \mathscr{G}\mid_{U_i})$. They're compatiable in the sense that for any open subset $W_{ij} \subset U_i \cap U_j$, we know

$$\mathrm{Res}_{U_i,U_i\cap U_j}\,\alpha_i = \mathrm{Res}_{U_j,U_i\cap U_j}\,\alpha_j \;\Rightarrow\; \alpha_i(W_{ij}) = \alpha_j(W_{ij}).$$

Now we try to define a natrual transformation $\alpha \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ such that $\alpha\mid_{U_i} = \alpha_i$. For any open subset $Y \subset U$, denotes $Y_i := Y \cap U_i$.

$$\begin{aligned}
\alpha(Y) : \mathscr{F}\mid_U(Y) &\to \mathscr{G}\mid_U(Y) \\
\mathscr{F}\mid_U(Y_i) \ni x &\mapsto \alpha_i(x).
\end{aligned}$$

This is map in sets, it's well-defined for $\{Y_i\}_{i\in I}$ is an open covering for Y and each α_i is compatiable. By construction we know $\alpha\mid_{U_i} = \alpha_i$. Hence we've checked gluability.

Need to check, but I think it's basically unwrapping a long long defintion

$\square$

18.1 Verification

See a post HERE.

Also see a lemma from Stacks Project HERE. This lemma basically proves gluability and uniqueness, based on the fact that Sheaf Hom is already a pre-sheaf. In the proof of the above lemma, we defined the natural transformation in the way such that the following diagram commute

$$\begin{tikzcd} s \in \mathscr{F}(U) \arrow[r, "\varphi_U"] \arrow[d] & \mathscr{G}(U) \arrow[d] \\ \mathscr{F}(V) \arrow[r, "\varphi_V"'] & \mathscr{G}(V) \end{tikzcd}$$

And this relies on the fact that $\mathscr{G}$ is a sheaf!, by looking at every $U \cap U_i$, which covers U.

18.2 Warning

Sheaf Hom does not commute with taking stalks. But there exists at least one map from

$$\begin{aligned} &\mathrm{Hom}(\mathscr{F}, \mathscr{G})_p \to \mathrm{Hom}(\mathscr{F}_p, \mathscr{G}_p) \\ &\{(\alpha, U) \ \mid \ p \in U, \ \alpha \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)\}/\sim_1 \mapsto ... \end{aligned}$$

18.3 Warning: References

See a post HERE, which contains a link to the detailed version of the counterexample HERE. According to post HERE, The direction was correct.

See Stefan's notes on Page 18 for a concrete example HERE. One comment on "Hom functor preserve limit" HERE.

Also, the first argument $\mathscr{F}$ doesn't have to be a sheaf, a pre-sheaf is enough. See a post HERE.

18.4 Counterexample

I'll add some details to the first example mentioned in the post linked above.

Proof.
For any $U \subset X$, where $\mathscr{F}$ is skycraper sheaf at $p \in X$ with value group A and $\mathscr{G}$ is a constant sheaf on topological space X with value group A. We claim that

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(U) = \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U) = 0$$

as an Abelian group for arbitrary $U \subset X$.

It suffices to check the above statment is correct when $U = X$ for $\mathcal{H}om(\mathscr{F}, \mathscr{G})$ is a sheaf, i.e. we need to show the group

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(X) = \mathrm{Mor}(\mathscr{F}, \mathscr{G}) = 0$$

Pick any natural transformation $\alpha \in \mathrm{Mor}(\mathscr{F}, \mathscr{G})$. We wish to show the group homomorphism $\alpha(U)$ is 0 (i.e. sends everything to 0 in the codomain $\mathscr{G}(U)$). For an open subset U, it admits an open covering $\{U_i\}$ in which every U_i is connected. While $\mathscr{G}$ is a sheaf, to prove $\alpha(U) = 0$ it suffices to check $\alpha(U_i) = 0$. Fix i and denote $U_i = U_0$. When $p \notin U_0$, then $\mathscr{F}(U_0) = 0$ and the map $\alpha(U_0) = 0$.

Now assume $p \notin U_0$. We can still argue $\alpha(U_0)$ is 0 group homomorphism by restrict it to a smaller open subset that doesn't contain p, because the skycraper sheaf will be 0 group.

Assume p is closed an not open, which means it's not isolated. Then $V := U \setminus \{p\}$ is an open subset that doesn't contain p, hence $\mathscr{F}(V) = 0$ by definition. ?

$$\begin{tikzcd}
\mathscr{F}(U_0) \arrow[r, "\alpha(U)"] \arrow[d, "\mathrm{Res}_{U_0 V}"'] & \mathscr{G}(U_0) = A \arrow[d] \\
\mathscr{F}(V) = 0 \arrow[r] & \mathscr{G}(V) = \oplus A
\end{tikzcd}$$

Notice that we assumed U_0 to connected, therefore $\mathscr{G}(U_0) = A$ is composed of single one copy of A. While $\mathscr{G}(V) = \oplus_{j \in J} A$ for some index set J. We claim that $\mathscr{G}(U_0) \to \mathscr{G}(V)$ is an injection:

- We claim that the map is in fact diagonal map by considering the following commutative diagram

$$\begin{tikzcd}
a \in \mathscr{G}(U_0) \arrow[r] \arrow[d, hook] & (a, a, ..., a)_{j \in J} \in \mathscr{G}(V) \arrow[d, hook] \\
\amalg_{q \in U_0} \mathscr{G}_q = \amalg_{q \in U_0} A & \amalg_{q \in V} A \arrow[l, hook']
\end{tikzcd}$$

  Because the map from $\mathscr{G}(U_0) \to \amalg_{q \in U_0} \mathscr{G}_q = \amalg_{q \in U_0} A$ is the diagonal map, therefore the map we considering must factor through it. Hence it could only be the diagonal map, with index set J.

Therefore by commutativity of the diagram together with the injectivity of the above map we can conclude $\alpha(U_0) = 0$ as expected.

$\square$

18.5 Abelian group structure

The Abelian group structure on $\mathcal{H}om(\mathscr{F}, \mathscr{G})(U)$ is given by defining

$$\alpha + \beta(U) : \mathscr{F}(U) \to \mathscr{G}(U) \ x \mapsto \alpha(U)x + \beta(U)x$$

for two natrual transformations $\alpha, \beta \in \mathcal{H}om(\mathscr{F}, \mathscr{G})(U)$. Notice that $\alpha + \beta$ is indeed a natural transformation, because it's compatible with restriction maps, which we could check by definition... Element $0 \in \mathcal{H}om(\mathscr{F}, \mathscr{G})(U)$ is just a natural transformation sends everything to $0 \in \mathscr{G}(U)$.

19 2.3.D.

19.1 (a)

Since the stalk doesn't commute with sheaf hom, I guess it could be more complicated to compute the stalk for sheaf hom $\mathcal{H}om$. So I'll check two functors are isomorphic in terms of *natural transformation*.

Let $\mathscr{C} := \underline{\{p\}}$ denotes the constant sheaf with value set $\{p\}$. Firstly, by definition we know for any open subset $U \subset X$

$$\mathcal{H}om(\mathscr{C}, \mathscr{F})(U) = \text{ Mor}(\mathscr{C}|_U, \mathscr{F}|_U).$$

Pick any element $\alpha \in \text{Mor}(\mathscr{C}|_U, \mathscr{F}|_U)$, which is a natural transformation.

Endow $\{p\}$ with discrete topology. For any open subset $V_1 \subset U$, we have set map $\alpha(V_1)$

$$\alpha(V_1) : \ \mathscr{C}|_U(V_1) = \mathscr{C}(V_1) \longrightarrow \mathscr{F}|_U(V_1) = \mathscr{F}(V_1).$$

The domain only contains one element: the set is all continuous map from some open subset in V_1 to $\{p\}$, hence there's only such function and we denote it as f. It follows that the map is completely determined by where f goes, $\alpha(V_1)$ could be identified with an element in $\mathscr{F}(V_1)$. In particular, for any $\alpha \in \text{Mor}(\mathscr{C}|_U, \mathscr{F}|_U)$, $\alpha \in \mathscr{F}(U)$, and

$$\mathcal{H}om(\mathscr{C}, \mathscr{F})(U) = \text{Mor}(\mathscr{C}|_U, \mathscr{F}|_U) = \mathscr{F}(U)$$

for any $U \subset X$, notice that it's in fact set-theorically bijection, hence it's a natural isomorphism and therefore two functors are isomorphic as expected.

Need to verify

19.2 References

A potentially related post HERE.

20 2.3.E.

- Diagram Chase: ?
- Universal Property: According to Definition 1.6.4., Kernel was defined using universal property. Therefore we can use universal property of $\text{Ker}(\phi(U))$ to deduce the unique existence of such map for $\text{Ker}(\phi(V)) \to \mathscr{G}(U) = 0$.

21 2.3.G. 2.3.H.

Notice that we're talking about pre-sheaves here. Hence these properties are satisfied by definition...

22 2.3.J.

Question: wouldn't it be better if $\underline{\mathbb{Z}}$ is replaced by $\underline{\mathbb{C}}$?

22.1 References

See a post HERE. In fact, the quotient sheaf could be not separated, see a post HERE.

See a detailed lecture note HERE.

And a solution HERE and HERE.

Here's another example where image presheaf might not be a sheaf.

23 2.4.A.

Proof.

$$\mathscr{F}(U) \to \prod_{p \in U} \mathscr{F}_p \tag{1}$$

$$g \mapsto (g_p, p \in U) \tag{2}$$

$$h \mapsto (h_p, p \in U) \tag{3}$$

Suppose we have $g_p = h_p$ for all points $p \in U$. By definition of germ, for any p, there exists open $W_p \subset U$ that contains p such that $g|_{W_p} = h|_{W_p}$. While $p \in U$ is arbitrary, $\{W_p\}$ forms an open cover for U. Since $\mathscr{F}$ is a separated presheaf, we may conclude $f = g \in \mathscr{F}(U)$. □

Remark 1. *See G ortz's book [**?**], page 52, Prop 2.23. (1); or Lemma 2.9. on Page 35 of [**?**].*

24 2.4.2. Compatible Germs

See a post regarding compatible germs and Espace Étalé.

24.1 Two Equivalent Definitions

The following two definitions are equivalent:

- *Element* $(s_p)_{p \in U} \in \prod_{p \in U} \mathscr{F}_p$ *consists of compatible germs* ...
- *There is an open covering* $\{U_i\}$ *of* U, ...

Proof.
Fix a point $p \in U$, pick some representative of $s_p \in \mathscr{F}_p$ as follow:

$$(\tilde{s}_p \in \mathscr{F}(U_p), \ldots)$$

where open subset $U_p \subset U$ contains p with $\tilde{s}_p \in \mathscr{F}(U_p)$. Then we take

$$\{U_p\}_{p \in U}$$

as an open covering of U and take corresponding sections $\tilde{s}_p \in \mathscr{F}(U_p)$ for $p \in U$. We can check conditions for the second statement is satisfied: Clearly $p \in U_p$ and s_p is the germ of $\tilde{s}_p$ at p for the first condition required this is true for any $q \in U_p$.

Conversely, pick any $p \in U$. We can find some U_i that contains p with a section $f_i \in \mathscr{F}(U_i)$ such that s_p is the germ of f_i at p. So we take

$$(f_i \in \mathscr{F}(U_i),\ U_p) \in \mathscr{F}_p$$

as representative of f_i. Notice that germ of f_i at any $q \in U_i$ is s_q as expected. □

24.2 Section induces a choice of compatible germs

Any sections s of $\mathscr{F}$ over U gives a choice of compatible germs for U.

Proof.
Let $s \in \mathscr{F}(U)$ as given and pick any open covering $\{U_i\}_{i \in I}$ of U where I is an index set. And define $f_i := s|_{U_i} \in \mathscr{F}(U_i)$. Now we may assume $p \in U_i$ for some U_i. The following diagram is commute for any $p \in U_i$, with element being the image under corresponding maps.

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r] & \mathscr{F}(U_i) \arrow[r] & \mathscr{F}_p \\
s \arrow[r, mapsto] \arrow[rr, bend right] & f_i := s|_{U_i} \arrow[r, mapsto] & s_p
\end{tikzcd}$$

Assume $p \in U_i$ for a fixed $i \in I$, then

$$s_p = (\mathscr{F}(U) \to \mathscr{F}_p)(s) = (\mathscr{F}(U_p) \to \mathscr{F}_p)(f_i)$$

by commutativity of the above diagram. Hence s_p is germ of f_i at p as expected. □

25 2.4.B.

We'll use equivalent definition of compatible germs to verify two sections are the same over intersections, which is done by Section 23.

Proof.
Fix $(s_p)_{p \in U}$ to be our choice of compatible germs. For $i = 1, 2$, we consider the following setup: $p_i \in U_i$ lies inside an open subset U_i where $f_i \in \mathscr{F}(U_i)$. Here $(f_i \in \mathscr{F}(U_i), ...)$ is the representative for f_i. Notice that we have $f_i|_p = s_p$: more

precisely, over $U_1 \cap U_2$, we have $f_1|_p = f_2|_p$. While according to Section 23, the morphism

$$\mathscr{F}(U_1 \cap U_2) \to \prod_{p \in U_1 \cap U_2} \mathscr{F}_p$$

is injective. Therefore we must have $f_1|_{U_1 \cap U_2} = f_2|_{U_1 \cap U_2}$. While $\mathscr{F}$ is a sheaf, Gluability Axiom concludes us with the existence of $f \in \mathscr{F}(U)$ such that $f|_{U_i} = f_i$; furthermore, we notice that $(f_p)_{p \in U} = (s_p)_{p \in U}$. □

$$\mathscr{F}(U) \to \prod_{p \in U} \mathscr{F}_p$$

In the last subsection of Section 24, we checked that any section will induce something called compatible germs on the left of the above map. Question is that what are the images of the above map? The following lemma will give us complete description of the image of the map above.

Proposition 1. *Any choice of compatible germs is the image of a section.*

Proof.
I'll take the second definitions of a choice of compatible germs: there is an open cover $\{U_i\}_{i \in I}$ of U where I is understood as an index set, and there are sections $f_i \in \mathscr{F}(U_i)$ such that s_p is the germ of f_i at any $p \in U_i$.

Let's consider two open subsets $U_i, U_j \subset U$ where $i, j \in I$ with $f_i \in \mathscr{F}(U_i), f_j \in \mathscr{F}(U_j)$. Assume $U_i \cap U_j \neq \emptyset$. Then for any $p \in U_i \cap U_j$, s_p is the germ of f_i and f_j at p, i.e.

$$[U_i, f_i] = s_p = [U_j, f_j] \in \mathscr{F}_p.$$

Hence there exists open subset $W_{ij,p} \subset U_i \cap U_j$ containing p such that $f_i|_{W_{ij,p}} = f_j|_{W_{ij,p}}$. Notice that $p \in U_i \cap U_j$ is arbitrary, therefore for any p there exists some $W_{ij,p}$ (depending on p). And all such $\{W_{ij,p}\}_{p \in U_i \cap U_j}$ will be an open covering for $U_i \cap U_j$, by Identity Axiom we know

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}.$$

While i, j were arbitrary, Gluability Axiom implies there exists $f \in \mathscr{F}(U)$ such that $f|_{U_i} = f_i$. And we can check that f is indeed the preimage of the given choice of compatible germs: pick some U_i containing p,

$$f_p = (\mathscr{F}(U_i) \to \mathscr{F}_p)(f|_{U_i})$$

Hence f_p is the germ of f_i at p.

□

26 2.4.C.

According to the hint, we have the following diagram commute for according to construction of 2.3.A.

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r, shift left, "\phi_1(U)"] \arrow[r, shift right, "\phi_2(U)"'] \arrow[d, "\pi"'] & \mathscr{G}(U) \arrow[d, hook, "\iota"] \\
\prod_{p\in U} \mathscr{F}_p \arrow[r, "\Phi"'] & \prod_{p\in U} \mathscr{G}_p
\end{tikzcd}$$

Hence we have

$$\iota \circ \phi_1(U) = \Phi \circ \pi = \iota \circ \phi_2(U) \;\Rightarrow\; \phi_1(U) = \phi_2(U)$$

for ι is injective.

27 2.4.D.

See Hartshorne [**?**] Chapter II, Proposition 1.1.

27.1 Warning

It could happen that two non-isomorphic sheaves have isomorphic stalks. See Sheaf Hom Section 18 regarding skycraper sheaf and constant sheaf.

See a post HERE.

28 2.4.E.

28.1 (a)

Pick a non-separated presheaf: See Stacks Project Definition 007A ?

Here is an example of gluable unseparated presheaf.

See a post discussing two examples where section isn't determined by stalks.

28.2 (b)

For Section 26, it's important to assume $\mathscr{G}$ is a sheaf.

See the answer by Andreas Blass HERE. For two pre-sheaves $\mathscr{F}, \mathscr{G}$, we might encounter a counterexample as follows:

Let $X = \{p, q\}$ be a discrete topological space with two points. Define two pre-sheaves $\mathscr{F}, \mathscr{G}$ on X with values in set $\{0, 1\}$. We specify two pre-sheaves as follow:

$U \subset X$	$\{p, q\}$	$\{p\}$	$\{q\}$	$\emptyset$
$\mathscr{F}(U)$	$\{0, 1\}$	$\{0\}$	$\{0\}$	$\{0\}$
$\mathscr{G}(U)$	$\{0, 1\}$	$\{1\}$	$\{1\}$	$\{1\}$

where the restriction maps for $\mathscr{F}$ and $\mathscr{G}$ are the only possible ones. Now we wish to define two different morphism of sheaves $\alpha, \beta : \mathscr{F} \to \mathscr{G}$ as natural transformations. For all open subsets except $U = \{p, q\}$, we have

$$\alpha(U) = \beta(U) : \{0\} \to \{1\}.$$

The place we intentionally make them different is when $U = \{p, q\}$:

$$\begin{aligned} \alpha(U) &: \{0,1\} \to \{0,1\} \quad \text{by} \quad 0 \mapsto 0,\ 1 \mapsto 1; \\ \beta(U) &: \{0,1\} \to \{0,1\} \quad \text{by} \quad 0 \mapsto 1,\ 1 \mapsto 0. \end{aligned}$$

Clearly as morphisms $\alpha \neq \beta$. But we have

$$\begin{aligned} \alpha_p \times \alpha_q : \prod_{x \in \{p,q\}} \mathscr{F}_x &\to \prod_{x \in \{p,q\}} \mathscr{G}_x \\ \{0\} \times \{0\} &\mapsto \{1\} \times \{1\}. \end{aligned}$$

But notice that there's only **one** choice of induced morphism between $\mathscr{F}_x \to \mathscr{G}_x$ for any $x \in X = \{p, q\}$. This means, even if we're given another different maps β, it will induce the same map on stalks.

?

28.3 (c)

Take a pre-sheaf $\mathscr{F}$ (on space X) that is not a sheaf. For example, constant pre-sheaf. We know the stalk of its sheafification at any $p \in X$ is

$$\mathscr{F}_p = \mathscr{F}_p^\sharp.$$

29 2.4.O.

A post HERE, HERE.

Another related post explaining isomorphic sheaves will give rise to injective and surjective maps on sections, this does not contradict the fact that surjectivity of morphism of sheaves cannot have analogus statement on sections.

See Richard on YouTube, Scheme Lecture 03 last example. Let $U = \mathbb{C} \setminus \{0\}$ with classical topology.

$$\mathscr{O}_X(U) \to \mathscr{O}_X^*(U)$$

is not surjective for $x \in \mathscr{O}_X^*(U)$ doesn't admits a holomorphic log for $f(0) = 0$ while exponential function can never be 0. Because we cannot find $logx$ as a holomorphic function on $\mathbb{C}$.

See a very detailed solution HERE.

29.1 Complex analysis background

See Theorem 6.1., 6.2. Chap 3 Page 98 [**?**].

See this great POST!

See a note HERE. A post HERE.

30 2.5.1.

30.1 Remarks

We can also define, for general open subset by taking limit: see comments of 009H by [**?**]; and [**?**] Proposition 2.20. on Page 51 with discussion above it.

31 2.5.B.

Proof.
... □

31.1 Another Proof

Proof.
A potentially wrong approach is to recognise that, in the definition of $\mathscr{F}(U)$: we set ?

$$\mathscr{F}(B) := \{(f_p \in F_p)_{p\in B} \ \mid \ \star\}.$$

The condition $\star$ is exactly the definition of compatible germs. While $F(B)$ could be identified with it's image in an injective map

$$F(B) \to \prod_{p\in B} F_p.$$

We proved in Section 25 that the image is precisely the compatible germs, therefore we must have they're isomorphic. □

31.2 Remarks

In fact, for sheaf on base F, the map

$$F(U) \to \prod_{p\in U} F_p$$

is injective. The proof is essentially the same as for sheaf on X.

32 2.5.D.

See [**?**] Chap 2 Exercise 2.7. on Page 44. This exercise will be used for Rising Sea [**?**] 4.3.B.

33 2.6.A.

I don't know what "natural" means. Is there a natural transformation?

I'll just check the isomorphism. And assume kernel sheaf has its corresponding the universal property.

Firstly, we have $(\mathrm{Ker}(\mathcal{F} \to \mathcal{G}))_p \to \mathrm{Ker}(\mathcal{F}_p \to \mathcal{G}_p)$ by universal property of $\mathrm{Ker}(\mathcal{F}_p \to \mathcal{G}_p)$. This is because we can compute the composition

$$(\mathrm{Ker}(\mathcal{F} \to \mathcal{G}))_p \to \mathcal{F}_p \to \mathcal{G}_p = 0.$$

For $s_p \in (\mathrm{Ker}(\mathcal{F} \to \mathcal{G}))_p$, we can diagram chasing it...

Conversely, ... a diagram chasing argument.

33.1 Solution

For details see Autumn HW2 2023 Problem 3. Or this online solution.

34 2.7.A.

Exercise 1. *Let $X = \{p, q\}, Y = \{w\}$ with discrete topolgoy. Let $\mathscr{G}$ be a constant sheaf on Y with value group A which contains at least two elements. Compute inverse image presheaf.*

Proof.
For any $U \subset X$, we have

$$\pi_{\mathrm{pre}}^{-1}\mathscr{F}(U) = \mathrm{Colim}_{V \supset \pi(U) = \{w\}} \mathscr{G}(V) = \mathrm{Colim}_{V = \{w\}} \mathscr{G}(V) = A.$$

Hence it's the inverse image pre-sheaf $\pi_{\mathrm{pre}}^{-1}\mathscr{F}$ is a constant pre-sheaf. It's not a sheaf for exactly the same reason constant pre-sheaf isn't a sheaf for failure of gluability. □ ?

Remark 2. *A post on inverse image presheaf.*

Also see the conversation here with Gemini.

35 2.7.D.

By definition... See [**?**] Definition on Page 65 of Section 2 of Chapter II.

36 2.7.G.

See **??**.

37 7.2.A.

MORPHISMS OF RINGED SPACES GLUE.

Proof.
For topological map f, the existence is clear (existence and the restriction will behave as desired). It remains to define the morphism of sheaves

$$f^\sharp : \mathscr{O}_Y \to \pi_* \mathscr{O}_X$$

and check that $(\pi|_{U_i})^\sharp : \mathscr{O}_Y \to (\pi|_{U_i})_* \mathscr{O}_{U_i} = \pi_i^\sharp : \mathscr{O}_Y \to \pi_{i,*}\mathscr{O}_{U_i}$.

By assumption, we have $\{\pi_i \ : \ U_i \to Y\}_{i\in\bullet}$ with $\pi_i^\sharp : \mathscr{O}_Y \to \pi_{i,*}\mathscr{O}_{U_i}$ for each $i \in \bullet$. Also recall that

$$\begin{aligned}(\pi_i|_{U_i\cap U_j})^\sharp : \mathscr{O}_Y \to (\pi_i|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j} = \\ (\pi_j|_{U_i\cap U_j})^\sharp : \mathscr{O}_Y \to (\pi_j|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j}.\end{aligned}$$

Therefore for any open subset $W \subset Y$, we have $(\pi_i|_{U_i\cap U_j})^\sharp(W) = (\pi_j|_{U_i\cap U_j})^\sharp(W)$. In particularly, we have

$$\begin{aligned}(\pi_i|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j}(W) &= \mathscr{O}_{U_i\cap U_j}((\pi_i|_{U_i\cap U_j})^{-1}(W)) \\ &= \mathscr{O}_{U_i\cap U_j}(\pi_i^{-1}(W) \cap U_i \cap U_j) \\ &= \mathscr{O}_X(\pi_i^{-1}(W) \cap U_i \cap U_j) \\ &= \mathscr{O}_X(\pi_j^{-1}(W) \cap U_i \cap U_j) \\ &= (\pi_j|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j}(W).\end{aligned}$$

Now we need to define $\pi^\sharp : \mathscr{O}_Y \to \pi_* \mathscr{O}_X$. For every open subset $W \subset Y$, we have to define a map

$$\pi^\sharp(W) : \mathscr{O}_Y(W) \to \pi_* \mathscr{O}_X(W).$$

For every $s \in \mathscr{O}_Y(W)$, we define(at this moment, we've constructed topologically the map π)

$$\begin{aligned}s_i := \pi_i^\sharp(W)(s) &= (\mathscr{O}_Y(W) \to \pi_{i,*}\mathscr{O}_{U_i}(W))(s) \\ &\in \mathscr{O}_X(\pi_i^{-1}(W) \cap U_i) = \mathscr{O}_X(\pi^{-1}(W) \cap U_i)\end{aligned}$$

We can check

$$\begin{aligned}s_i|_{\pi^{-1}(W)\cap U_i\cap \pi^{-1}(W)\cap U_j} &= (\mathscr{O}_X(\pi^{-1}(W) \cap U_i) \to \mathscr{O}_X(\pi^{-1}(W) \cap U_i \cap U_j))(s_i) \\ &= (\mathscr{O}_X(\pi_i^{-1}(W) \cap U_i) \to \mathscr{O}_X(\pi_i^{-1}(W) \cap U_i \cap U_j))(s_i) \\ &= (\mathscr{O}_Y(W) \to \mathscr{O}_X(\pi^{-1}(W)) \to \mathscr{O}_X(...))(s) \\ &= (\mathscr{O}_X(\pi_j^{-1}(W) \cap U_j) \to \mathscr{O}_X(\pi_j^{-1}(W) \cap U_i \cap U_j))(s_j) \\ &= s_j|_{\pi^{-1}(W)\cap U_i\cap \pi^{-1}(W)\cap U_j}\end{aligned}$$

Notice that we have $\{\pi^{-1}(W) \cap U_i\}_{i\in\bullet}$ covers $\pi^{-1}(W)$ and $\mathscr{O}_X$ is a sheaf. Therefore there exist $S \in \mathscr{O}_X(\pi^{-1}(W)) = \pi_* \mathscr{O}_X(W)$ such that $S|_{U_i} = s_i$. Hence for every $W \subset Y$ we've defined the map $\pi^\sharp$. And we can check that for all open subsets $W \subset U_i$,

$$(\pi|_{U_i})^\sharp(W) = \pi_i^\sharp(W) \Rightarrow (\pi|_{U_i})^\sharp = \pi_i^\sharp.$$

□ ???

37.1 Comments

See Prop 3.5. [**?**] on Page 70. For details, see "Glueing morphisms of sheaves together - can I just do this?" and "morphism of ringed spaces glue".

38 7.2.D.

Exercise 2 (Pulling Back $\mathscr{O}$-module)**.** *Suppose $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a morphism of ringed spaces. A slight variation on the inverse image allows us to pull back an $\mathscr{O}_Y$-moudle $\mathscr{G}$ to get an $\mathscr{O}_X$-module $\pi^*\mathscr{G}$.*

38.1 (a)

$$(X, \mathscr{O}_X) \xrightarrow{\pi_1} (X, \pi^{-1}\mathscr{O}_Y) \xrightarrow{\pi_2} (Y, \mathscr{O}_Y) \tag{4}$$

With the notation displayed as above, we need to firstly define π_1, π_2 and then check the commutativity. Both definition and checking commutativity involves both the topological maps as well as pullback of corresponding sheaves.

1. Now we'll define two morphisms

   (a) For π_1. We take the identity map id_X over X to be the topological map; it remains to define

   $$\pi^{-1}\mathscr{O}_Y \to \pi_{1,*}\mathscr{O}_X$$

   Fix $U \subset X$, and we let $V_1 \supset V_2 \supset \pi(U)$ inside Y.

   $$\begin{tikzcd}
   \mathscr{O}_Y(V_1) \arrow[r] \arrow[d] & \mathscr{O}_Y(V_2) \arrow[r] \arrow[d] & \pi^{-1}\mathscr{O}_Y(U) = \mathsf{Colim}_{V \supset \pi(U)} \mathscr{O}_Y(V) \arrow[d] \\
   \mathscr{O}_X(\pi^{-1}(V_1)) \arrow[r] & \mathscr{O}_X(\pi^{-1}(V_2)) \arrow[r] & \pi_{1,*}\mathscr{O}_X(U) = \mathscr{O}_X(U)
   \end{tikzcd}$$

   (b) For π_2. We take π to be the topological map. Hence we would like to unwrap the definition of

   $$\mathscr{O}_Y \to \pi_{2,*}\pi^{-1}\mathscr{O}_Y. \tag{5}$$

   For $W \subset Y$, we simply use the canonical map to the colimit (induced by the universal property of colimit)

   $$\begin{aligned}
   \mathscr{O}_Y(W) &\to \pi_{2,*}\pi^{-1}\mathscr{O}_Y(W) && (6)\\
   &= \pi^{-1}\mathscr{O}_Y(\pi^{-1}(W)) && (7)\\
   &= \mathop{\mathsf{Colim}}_{V \supset \pi(\pi^{-1}(W))} \mathscr{O}_Y(V) && (8)\\
   &= \mathop{\mathsf{Colim}}_{V \supset W} \mathscr{O}_Y(V). && (9)
   \end{aligned}$$

2. We check the commutativity. For topological space π_1, π_2 together work well. For the sheaf morphism, we intend to check two morphisms are isomorphic:

$$\phi_1 : \mathscr{O}_Y \to \pi_* \mathscr{O}_X = \tag{10}$$
$$\phi_2 : \mathscr{O}_Y \to \pi_{2,*}(\pi^{-1}(\mathscr{O}_Y)) \to \pi_{2,*}\pi_{1,*}\mathscr{O}_X. \tag{11}$$

We could do this by checking for arbitrary $W \subset Y$,

$$\phi_1(W) = \mathscr{O}_Y(W) \to \pi_* \mathscr{O}_X(W) \tag{12}$$
$$\phi_2(W) = \mathscr{O}_Y(W) \to \ldots \to \pi_* \mathscr{O}_X(W). \tag{13}$$

Therefore we obtained the commutativity for the sheaves.

Remark 3. *For composition of morphism in ringed spaces, see this post.*

39 7.2.E.

See Hartshorne [**?**] Page 72 for details.

40 7.3.A.

For a local homomorphism, we have

$$\mathfrak{m}_q \subset \mathscr{O}_{Y,q} \to \mathscr{O}_{X,p} \to \kappa(p).$$

It will induce a map $\kappa(q) \to \kappa(p)$.

41 7.3.B.

42 7.3.D.

!

43 7.3.F.

- See this conversation with Gemini.
- See Remark 3.15. [**?**]

44 7.3.G.

See [**?**] Proposition 3.4. Chap 3 on Page 69.
See Tag 01I1.

45 7.3.6.

For "fun question": is that maximalSpec?

46 7.3.O.

See conversation with Gemini.

47 7.3.L.

Nice hint. See Hartshorne Chap 2 Exercise 2.7. [**?**].

48 7.4.

Skipped...

49 7.5.B.

Tag 0CC1

50 7.5.4.

Gathmann's notes Definition 9.5 provided the proof, with one subtlety missed. See Gemini: Dominant Maps and Rational Map Composition

51 14.1.C.

Taking possible intersections of the covers for which both locally free sheaves are free, we may reduce the problem to check

$$(\mathcal{O}^{\otimes mn})|_{U_i} \cong \mathcal{H}om(\mathcal{F}, \mathcal{G})|_{U_i}.$$

And this is a direct consequence of the following observation: Page 179, equatioin (7.4.11) in [**?**].

52 14.1.D.

Proof.
In this proof, we'll utilise the following algebraic fact: For any ring R and any R-module M, we have an isomorphism

$$M \cong M^{\vee\vee}.$$

In fact, this solves the problem over affine schemes. Applying this fact repeatedly, we've defined the morphisms over an open cover of our domain. Thus we only need to check if the morphism glues together. Over the intersection, they agree because localisation commutes with dual functor (i.e. the hom functor). □

Proof.
In this attempt, we'll use transition maps to define our $\mathscr{E}^\vee$. Let $\{U_i\}$ be an open cover of X such that $\psi_i : \mathscr{E}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus n}$. Then our transition maps are given by

$$T_{ij} := \psi_i \circ \psi_j^{-1} \in \mathsf{GL}_n(\mathcal{O}_X(U_i \cap U_j)).$$

Now we need to examine the dual sheaf $\mathscr{E}^\vee$. Over U_i, we need a suitable choice for the isomorphism

$$\varphi_i : \mathscr{E}^\vee|_{U_i} \to \mathcal{O}_{U_i}^{\oplus n} \cong (\mathcal{O}_{U_i}^{\oplus n})^\vee. \tag{14}$$

And our choise will be $\varphi_i := \psi_i^{\vee,-1}$. Therefore our new transition maps for $\mathscr{E}^\vee$ are given

$$\begin{aligned} S_{ij} &= \varphi_i \circ \varphi_j^{-1} && (15)\\ &= (\psi_i^{\vee,-1}) \circ (\psi_j^{\vee,-1})^{-1} && (16)\\ &= (\psi_i^{\vee,-1}) \circ \psi_j^\vee && (17)\\ &= (\psi_j \circ \psi_i^{-1})^\vee && (18)\\ &= T_{ij}^{-1,\vee}. && (19) \end{aligned}$$

Thus applying twice the dual functor, the transition maps will restore the original T_{ij}, and we are done. □

53 14.2.A.

My discussion with ChatGPT HERE
See this POST for an easier solution.

54 14.4.D.

Exercise 3. *Suppose $\pi : X \to Y$ is a finite morphism of locally Noetherian schemes. If $\mathscr{F}$ is a coherent sheaf on X, show that $\pi_*\mathscr{F}$ is a coherent sheaf. Hint: Show first that $\pi_*\mathscr{O}_X$ is finite type.*

Proof.

1. In the following, we'll check $\pi_*\mathscr{O}_X$ is finite type. While being finite type satisfies the requirement of Affine Communication Lemma, it suffices to show the existence of a cover of affine opens $\{U_i\}_{i\in I}$ such that

$$\pi_*\mathscr{O}_X(U_i)$$

is finitely generated. In fact, we may take $\{U_i\}_{i\in I}$ to be a cover of open affines inside Y. Recall the morphism π is finite: in the textbook, it's automatically affine; hence we may assume $\pi^{-1}(U_i) = V_i$ where V_i is affine.

According to the definition of finite morphism, we have the following exact sequence

$$\mathscr{O}_Y(U_i)^{\oplus n} \to \pi_*\mathscr{O}_X(U_i) = \mathscr{O}_X(\pi^{-1}(U)) = \mathscr{O}_X(V_i) \to 0. \tag{20}$$

for some choice of $n \in \mathbb{Z}$. So we may conclude $\pi_*\mathscr{O}_X$ is of finite type.

2. Since $\mathscr{F}$ is a coherent sheaf, under the locally Noetherian assumption, it is of finite type. Thus over U_i, the following is exact

$$\mathscr{O}_X|_{U_i}^{\oplus l} \to \mathscr{F}|_{U_i} \to 0. \tag{21}$$

In general, the functor π_* is only left-exact. One important fact is that π is finite thus an affine morphism, so apply π_* will be right-exact. Therefore we have the following exact.

$$\pi_*(\mathscr{O}_X|_{U_i}^{\oplus l}) \to \pi_*\mathscr{F}|_{U_i} \to 0. \tag{22}$$

On the other hand, the last proof gave us

$$\mathscr{O}_Y|_{U_i}^{\oplus m} \to \pi_*\mathscr{O}_X|_{U_i} \to 0. \tag{23}$$

Combining two exact sequences above, we may conclude over U_i,

$$(\mathscr{O}_Y|_{U_i})^{\oplus ml} \to \pi_*\mathscr{F}|_{U_i} \to 0 \tag{24}$$

is exact. Therefore $\pi_*\mathscr{F}$ is of finite type.

□

Remark 4. *In fact, the standard way to solve the problem is to translate everything into commutative algebra. Relevant algebraic fact we'll use is the transitivity of f.g. module. See the conversation with Gemini.*

14.5.L.

54.1 (a)

- Used 14.5.I. when simplify the expression.

- And use adjointness between π^* and π_*.
- We CAN form tensor product of morphisms of sheaves: this can be done by recalling the universal property of sheafification.
- See conversation with Gemini.

55 15.2.J.

Remark 5.

56 18.5.A.

56.1 (a)

Assume (b).

$$\begin{aligned} h^0(C, \omega_C) &= h^0(C, \omega_C \otimes \omega_C^\vee) + 1 - g + \deg \omega_C \\ &= h^0(C, \mathcal{O}_C) + 1 - g + 2g - 2 \\ &= 1 + 1 - g + 2g - 2 = g. \end{aligned}$$

56.2 (b)

See Chap IV Example 1.3.3. [**?**].

57 23.1.C.

Almost by definition...

57.1 References

See this POST, and a detailed POST.
For details, see [**?**] Theorem 7.2. on Page 405.

58 23.4.F.

See remarks of **??**.